Patterns of System Strain: Crowding and Reliability in the New York City Subway System

Prabhleen Gill
School of Engineering and Computer Science
University of the Pacific
Stockton, USA
p_gill9u.pacific.edu

***Abstract*—This project analyzes congestion and reliability challenges in the New York City subway system using publicly available Metropolitan Transportation Authority (MTA) data. Two aspects of system performance are examined: ridership patterns as an indicator of crowding and operational delay data as a measure of service reliability. Crowding was analyzed using ridership trends and a crowding index to identify differences in demand intensity across stations and times. Reliability was assessed by estimating train delays through comparisons of scheduled and real-time operational data. Results show that subway demand is concentrated during peak commuting periods, while crowding patterns vary meaningfully across the system. Delay analysis identifies differences in reliability across lines and stations, with infrastructure and equipment issues emerging as a leading reported cause of disruptions. Together, these findings highlight how passenger demand and operational constraints contribute to system strain and demonstrate how transit data can be used to better understand congestion and reliability challenges in urban transportation systems.**

***Index Terms*—Public Transport, Subway Reliability, Transit Delays, Crowding**

I. Introduction

The New York City Subway is one of the largest and most complex urban transit systems in the world, serving approximately 3.376 million riders on an average weekday in 2024 [1]. For many New Yorkers, particularly those in outer boroughs with limited access to alternative transportation, the subway is not only a convenience but also a necessity. Hence, the performance of the subway system has direct consequences for the daily lives and economic mobility of millions of residents.

Despite its scale and importance, the NYC subway system has long struggled with reliability and capacity. Decades of aging infrastructure, such as legacy signal systems, and high passenger demand have placed significant strain on the system [2]. In recent years, the MTA has undertaken substantial capital investment efforts to modernize the network. According to the 2020-2024 Capital Plan, approximately $33.2 billion was allocated for New York City Transit [3]. Yet challenges related to crowding and service disruptions remain persistent, raising questions about how demand pressures and operational reliability continue to shape rider experience.

This study examines congestion and reliability challenges in the New York City subway system through two complementary perspectives: ridership patterns as an indicator of crowding and operational delay data as a measure of service reliability. Using publicly available MTA data, this analysis explores how demand and delays vary across time, stations, and line groups, with the goal of identifying patterns of system strain and better understanding the factors that shape subway performance.

II. Literature Review

Crowding in urban public transportation systems has become a critical issue in major metropolitan areas, affecting both system performance and passenger experience. As transit demand increases, understanding and mitigating crowding has emerged as an important area of research. Existing literature spans methodological approaches for estimating crowding, analyses of passenger behavior and costs, and empirical studies across different transit systems.

A significant body of work focuses on leveraging data to estimate and model crowding levels in transit systems. Caspari et al. [7] proposes a real-time approach for estimating platform crowding in the New York City subway, demonstrating the potential of integrating operational data to monitor congestion dynamically. Similarly, Ceapa et al. [8] analyzes congestion patterns in the London Underground using trip data, highlighting temporal and spatial variations in station crowding.

Smart card data has been widely used as a foundation for demand modeling. Chu et al. [9] enhances archived transaction data to improve transit demand estimation, while Jang [14] utilizes such data to examine travel times and transfer behavior. When combined, these studies demonstrate that large-scale passenger data enables detailed insights into system usage.

Researchers have examined the impacts of crowding on passengers and system efficiency. Tirachini et al. [16] provides a comprehensive overview of how crowding affects user behavior, operational performance, and demand estimation. Similarly, Douglas et al. [11] quantifies the cost of overcrowding, emphasizing its implications for transit planning. Recent work has also explored the subjective experience of crowding. Daziano et al. [10] uses visual information to estimate how passengers perceive and value crowded spaces in the NYC subway. Haywood et al. [13] further investigates how crowding costs are distributed among passengers and why individuals respond differently to congestion. These studies highlight that crowding is not only a physical constraint but also a behavioral and economic phenomenon, influencing route choice, willingness to pay, and overall satisfaction.

Empirical studies across different cities provide additional context for understanding crowding dynamics. Lam et al. [15] examines crowding effects in Hong Kong's light rail system, identifying key operational challenges associated with high passenger density. Research on the Paris transit system by Haywood et al. [12] reveals inequalities in how crowding costs are experienced across populations, while Ceapa et al. [8] highlights similar issues in London. These international studies suggest that while crowding is a universal issue, its causes and impacts vary depending on system design, passenger demographics, and operational strategies.

III. Methodology

A. Data Sources

Subway ridership data was obtained from the MTA Subway Hourly Ridership dataset [4]. It included information such as date, station ID and name, borough, fare class category, ridership, and transfers. Ridership counts for a station included both transferring and non-transferring passengers. Staten Island was excluded from analysis because the Staten Island Railway operates separately from the primary subway network and was outside the scope of this study. Because the dataset contained nearly 16 million rows, analysis was conducted using DuckDB queries. Preliminary exploratory analysis was used to examine relationships among variables and analyze station-level ridership activity.

Static GTFS data from MTA developer resources was used to obtain schedule-related information, including trips, stops, routes, and stop times [5]. This data was used both to estimate trip frequency for constructing the crowding index and to provide scheduled arrival and departure times for delay calculations.

Real-time trip data was extracted from Subway Data NYC [7], an open source archive of historical subway feeds collected through MTA API endpoints. This data included the features trip ID, stop ID, arrival and departure times, last observed time, and marked past time. For preprocessing, arrival and departure times were converted from Unix time to local New York time, and the last observed time and marked past time features were dropped. Directional suffixes (N/S) on the stop identifiers were removed to standardize station matching.

Additional station reference information was obtained from the MTA Subway Station dataset, which included station identifiers, GTFS stop identifiers, and related station metadata [6]. This dataset was used as a lookup table to map stop identifiers in the real-time trip data to station names.

B. Crowding Measurement

This study examines crowding and delay as complementary dimensions of subway performance at the station and line level across the New York subway system. Crowding was approximated using a crowding index based on riders per scheduled trip at each station. Raw ridership alone is insufficient as a measure of crowding because the number of scheduled trips varies across stations. For example, a station with high ridership and low number of scheduled trips would be considered more crowded than a station with high ridership and a higher number of scheduled trips.

The crowding index for each station was calculated as the ridership over the scheduled trips, where ridership was obtained from the MTA Subway Hourly Ridership Dataset [4] and scheduled trips were derived from the GTFS data [5]. Because crowding values varied substantially across stations and time periods, a logarithmic transformation was applied to reduce skewness and improve comparability across observations.

C. Delay Measurement

Delay is defined as the difference, in seconds, between scheduled and actual arrival and departure times. Arrival time refers to the time at which a train reaches a station, while departure time refers to the time at which a train leaves a station. To maintain consistency across service patterns, delays were calculated only for weekday trips operating from 6 AM to 12 AM. Because stations differ in service characteristics and many are associated with multiple routes, stations were categorized by their line color. Color categories are shown in Table 1.

TABLE I: Subway Line Color Categories

Color	Lines
Red	1, 2, 3
Green	4, 5, 6
Blue	A, C, E
Orange	B, D, F, M
Yellow	N, Q, R, W
Purple	7
Brown	J, Z
Gray	L
Light Green	G
Shuttle	S

It is important to note that certain subway stations operate on more than one line color. To avoid misattributing delays at multi-line stations, color-based analysis was restricted to stations associated with a single line color. Because stations were measured based on their GTFS Stop ID rather than name, stations with unique IDs for each line color were still included. For example, Times Sq-42 St covers the red, yellow, purple, and shuttle lines, and the respective GTFS Stop IDs for each color are 127, R16, 725, and 902. Woodhaven Blvd covers the brown, orange, and yellow lines, but has only two GTFS Stop IDs: J15 for brown and G11 for orange and yellow. Because G11 represents two different line colors, its trips were not included in the delay calculation and analysis. The real-time data used for retrieving actual trip times did not provide information on the specific line a trip occurred.

Delays were computed as the difference between matched scheduled and actual arrival and departure times. Scheduled times were obtained from the GTFS data, and actual times were obtained from the real-time trip information extracted from Subway Data NYC. Because real-time trip records were stored as daily files, a Python pipeline was developed to iterate through files, preprocess both datasets, perform temporal matching, calculate delays, and store summary statistics.

Preprocessing included converting Unix timestamps in the real-time data and GTFS data into localized New York timestamps, standardizing stop identifiers, excluding overnight trips between midnight and 6 AM, and labeling trips by line color. Additionally, directional suffixes (N/S) were removed from the stop IDs to standardize station matching. Because trip identifiers were not directly compatible across the two datasets, scheduled and actual trip times were matched using nearest-neighbor temporal alignment implemented with the pandas merge_asof() method, with tolerance set to 20 minutes to capture schedule matches while limiting implausible pairings. To avoid misattributing delays at stations serving multiple line colors, color-based analysis was restricted to stations associated with a single line color, while stations with distinct GTFS stop IDs for separate line colors were retained. Delays were then analyzed separately by line color. For each line group and day, average arrival and departure delays were computed, along with the station exhibiting the highest average delay. These daily summaries were aggregated into a dataframe for comparing reliability across line groups and identifying recurring bottleneck stations. To reduce the influence of implausible matches and extreme values, delays exceeding one hour were removed prior to additional outlier filtering using interquartile range fences. This filtering was used to improve the stability of average delay estimates.

IV. Results

A. Ridership Patterns

Ridership patterns were first examined using the MTA Subway Hourly Ridership dataset to identify temporal and spatial demand trends during the study period. Monthly ridership from April to September 2025, as shown in Figure 1, showed relatively stable demand, with ridership highest in May and lowest in August.

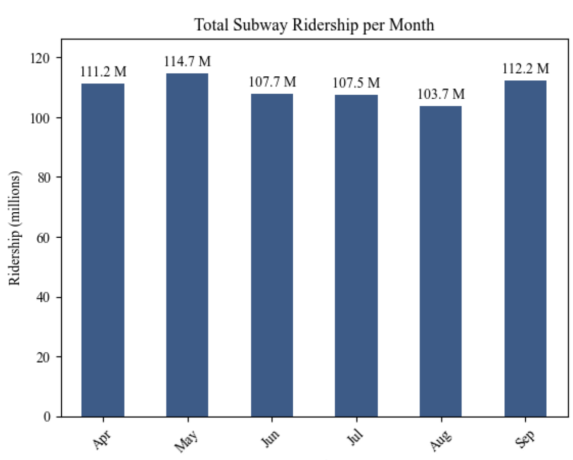


Fig. 1: Monthly subway ridership.

Figure 2 shows the total ridership for each New York City borough. Ridership was highly concentrated in Manhattan, with about 365.2 million riders in the six-month period, substantially exceeding the other boroughs. Figure 3 displays the top 20 stations by ridership. Times Sq-42 St was the station with the highest total ridership from April to September 2025. This station is the largest subway station complex in the NYC Subway system, serving five different lines. Grand Central-42 St had the second highest total ridership, and it serves three different lines. A large majority of the stations in this figure are located within Manhattan, indicating this borough is more prone to experience concentration of demand at major hubs.

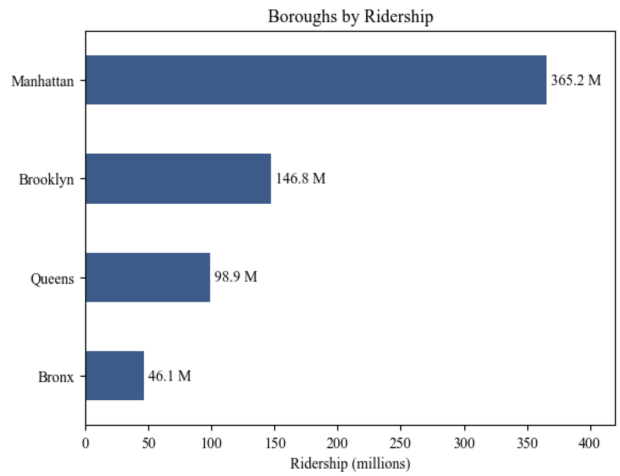


Fig. 2: Boroughs by ridership.

Figure 4 depicts the daily subway ridership for each month in the study period. Trends show that weekends have significantly lower riderships compared to weekdays, with Sundays being the lowest. Among the weekdays, ridership is highest from Tuesday to Thursday, with Wednesday having slightly more riders. September 11, 2025 was the day with the highest ridership in this study period, with about 4.55 million total riders.

Figure 5 depicts the hourly subway riderships for each month in the study period. There were consistent demand peaks during commuting periods across all months. Ridership was lowest during early morning hours, increased sharply during morning commute periods, remained elevated throughout daytime hours, and declined following the evening peak. While patterns were consistent across all months, minor seasonal variation was observed, including slightly lower morning ridership hours during summer months.

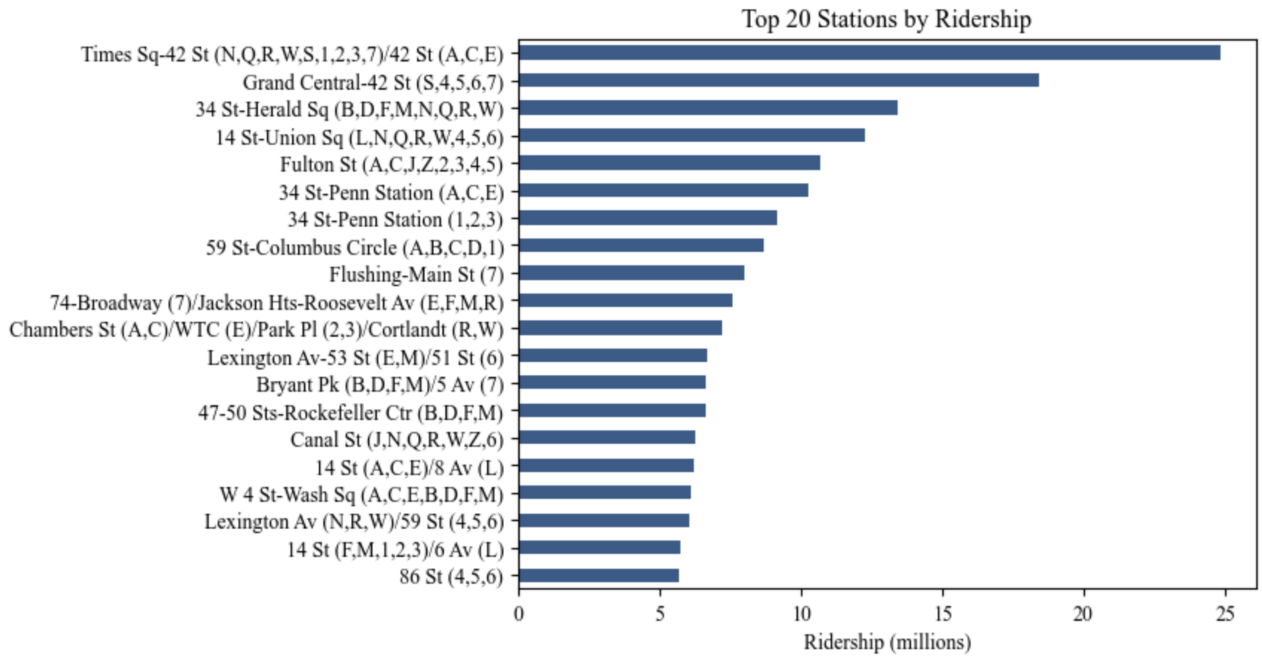


Fig. 3: Top 20 stations by ridership.

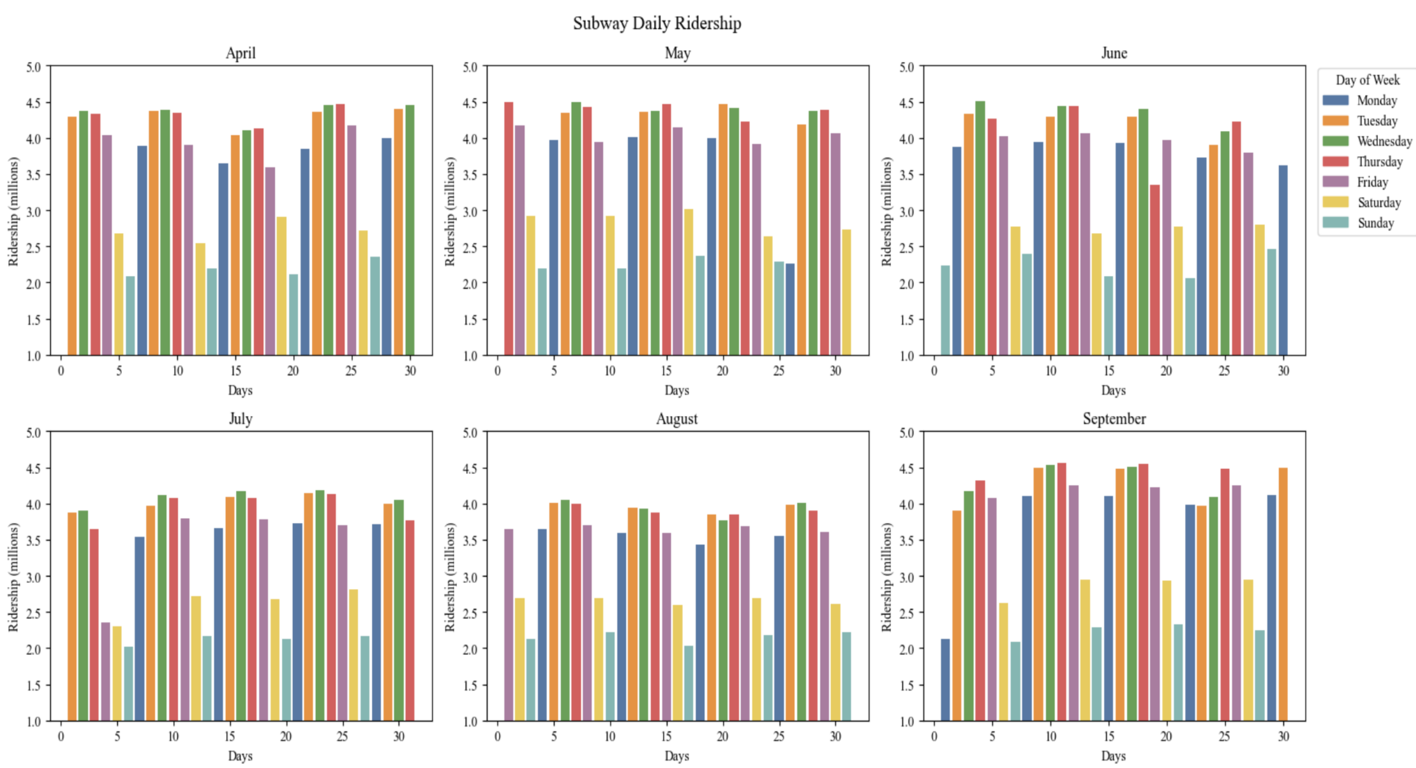


Fig. 4: Daily subway ridership by month.

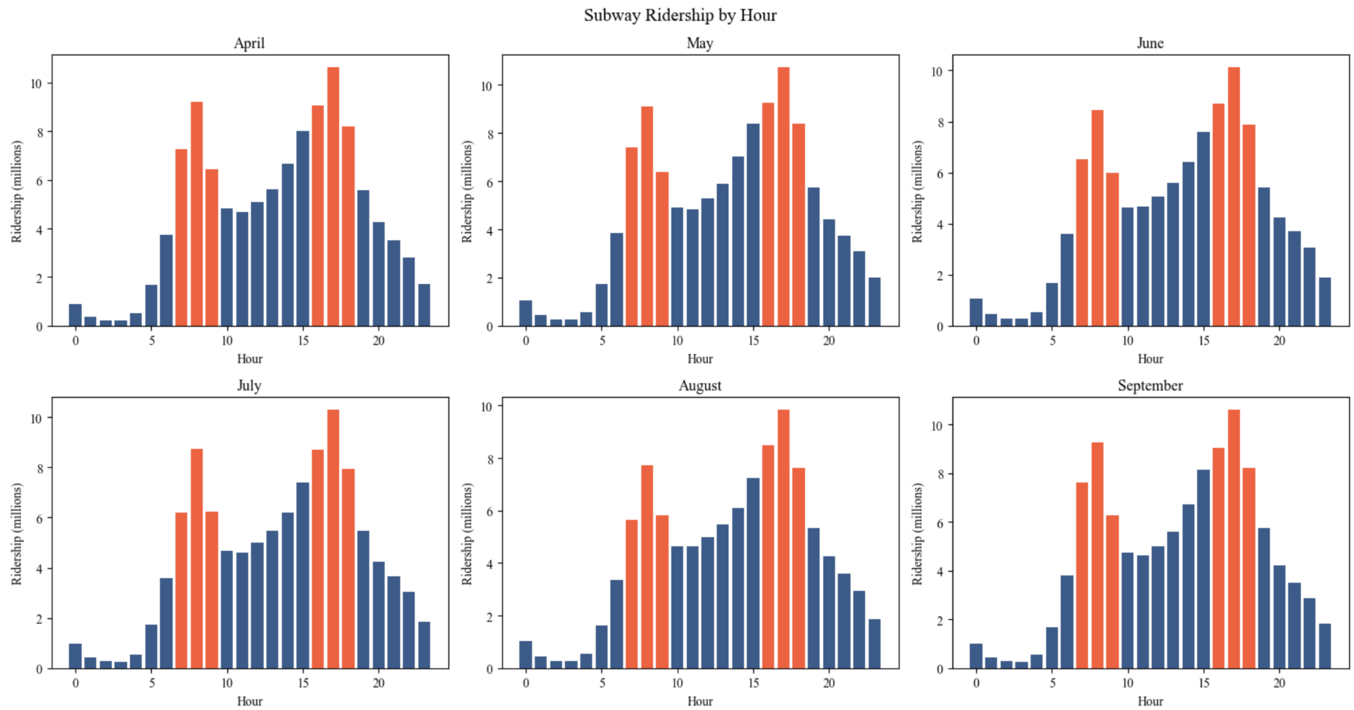


Fig. 5: Hourly subway ridership by month.

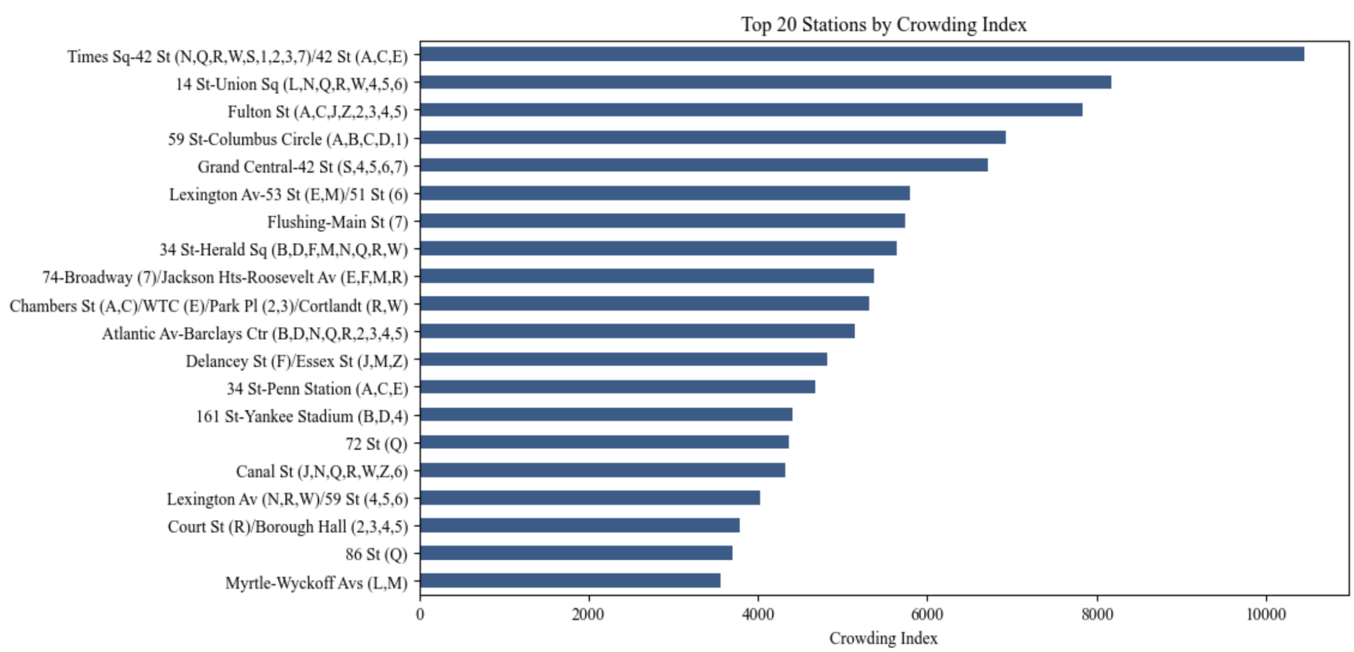


Fig. 6: Top 20 stations by crowding index.

B. Crowding Analysis

Figure 6 shows the top 20 stations with the highest crowding index. Times Sq-42 St had the highest crowding index, followed by 14 St-Union Sq and Fulton St. It can be observed that total ridership does not necessarily equate to the level of crowding. As observed in Figure 3, Fulton St was the station with the fifth highest ridership, though its crowding index was higher than those of 34 St-Herald Sq and Grand Central-42 St. The two stations ranked slightly lower on the crowding index scale, at 8th and 4th respectively. This suggests that station ridership alone does not fully capture crowding conditions and that the relationship between service frequency and demand plays an important role.

Figure 7 depicts the distribution of crowding indices on a logarithmic scale. The distribution is largely concentrated at indices 6 to 8, with a peak near 6.5. This suggests that crowding levels are clustered around a typical range rather than being uniformly distributed across stations. While some variation exists, the log-transformed distribution indicates that differences in crowding intensity are less extreme than raw values may suggest. The distribution also suggests that the demand-to-service ratios are similar across stations, which may imply that service frequency often increases with demand at most stations.

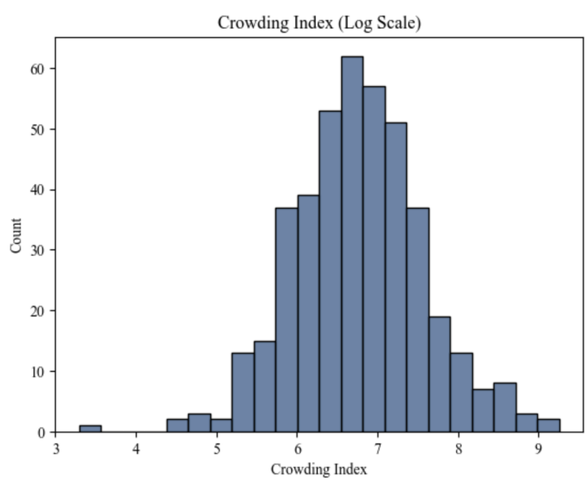


Fig. 7: Crowding index on the logarithmic scale

A key result of this analysis is that ridership alone is an incomplete indicator of crowding. Incorporation service frequency through the crowding index reveals differences in congestion that are not apparent from passenger volume alone.

C. Delay Analysis

Because the light-green and shuttle line groups had limited representation and produced unstable estimates, they were excluded from line-level delay comparisons.

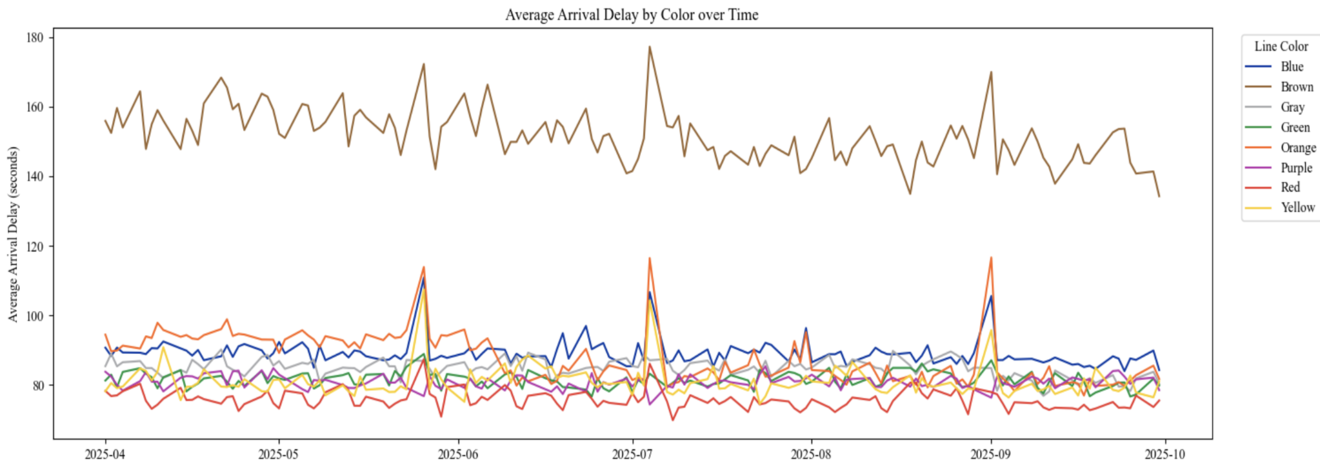
Figure 8 shows the average arrival and departure delays for each line color over time. On average, the brown line consistently experienced higher average delays than the other lines for both arrivals and departures. Several lines, including the brown, blue, orange, and yellow lines, exhibited noticeable increases in delays during June, July, and September. Similar patterns were observed for the red and green lines to a lesser extent, while the gray and purple line did not follow this trend. These increases in delay levels during the summer months align with the higher incidence of reported disruptions during this period, as shown in Figure 10.

While the brown line maintained the highest average delays, differences between arrival and departure delays were observed across lines. The purple and gray lines exhibited higher average departure delays compared to arrival delays, as shown in Figure 8. Furthermore, several higher-ridership lines, such as red, yellow, and green, exhibited lower average delays than some lower-ridership lines. This suggests that higher demand does not necessarily correspond to worse reliability and that the relationship between ridership and delays is not straightforward.

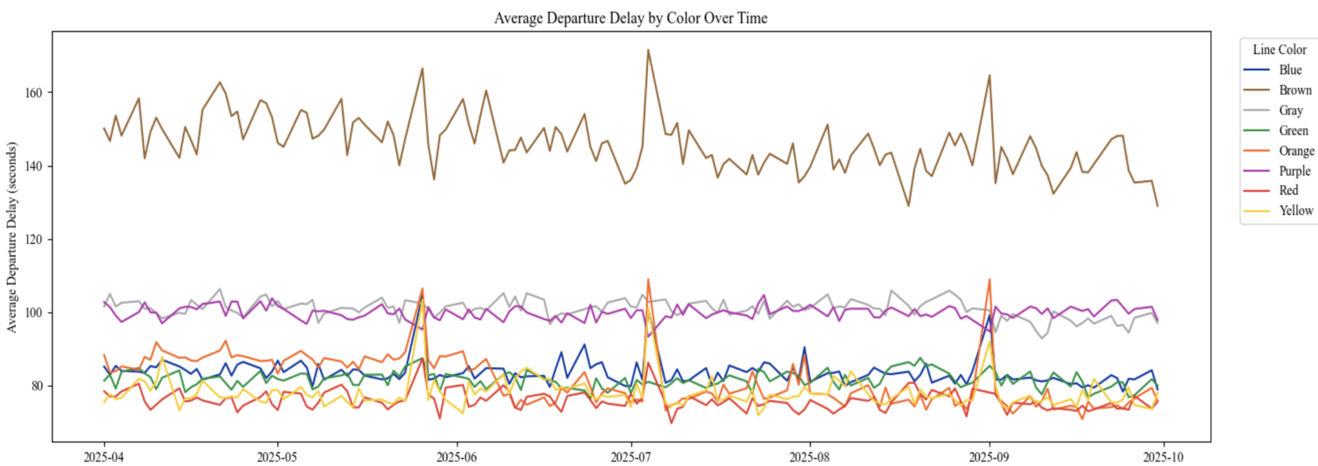
Figure 9 displays the top 20 stations by worst average arrival and departure delays. A station is considered the worst for a given day if it has the highest average delay among all stations on that day. According to the figure, WTC Cortlandt was identified as the worst-delay for 126 out of the 182 days in the study period, accounting for nearly 70% of all days. This indicates a strong concentration of delay severity at this station. It was followed by 52 St and 14 St-Union Sq, which also appeared frequently among worst-delay stations. Further analysis shows that delay patterns vary by line within multi-line stations. For example, although 14 St-Union Sq serves multiple lines, it ranked most prominently as a worst-delay station on its gray line, while its yellow and green lines did not appear as frequently among the rankings. This suggests that delays at multi-line stations may be driven by specific line operations rather than station-level demand alone.

Differences in arrival and departure delay patterns were also observed at the station level. Certain stations, such as 86 St, 57 St, and Beach 90 St, ranked highly in worst average arrival delays but did not appear among worst average departure delays, while others such as Roosevelt Island showed the opposite pattern. These differences indicate that delay behavior may vary depending on whether trains are arriving at or departing from a station.

A key result of the delay analysis is that stations with the most severe delays generally do not coincide with the highest-ridership stations identified in Figure 3. Only a small number of stations, such as 14 St-Union Sq and 86 St, appeared in both the top ridership and worst-delay rankings. This suggests that delay severity may be driven less by overall demand levels and more by localized operational and infrastructure conditions.



(a) Average arrival delays by line color



(b) Average departure delays by line color

Fig. 8: Average delays by subway line color over time.

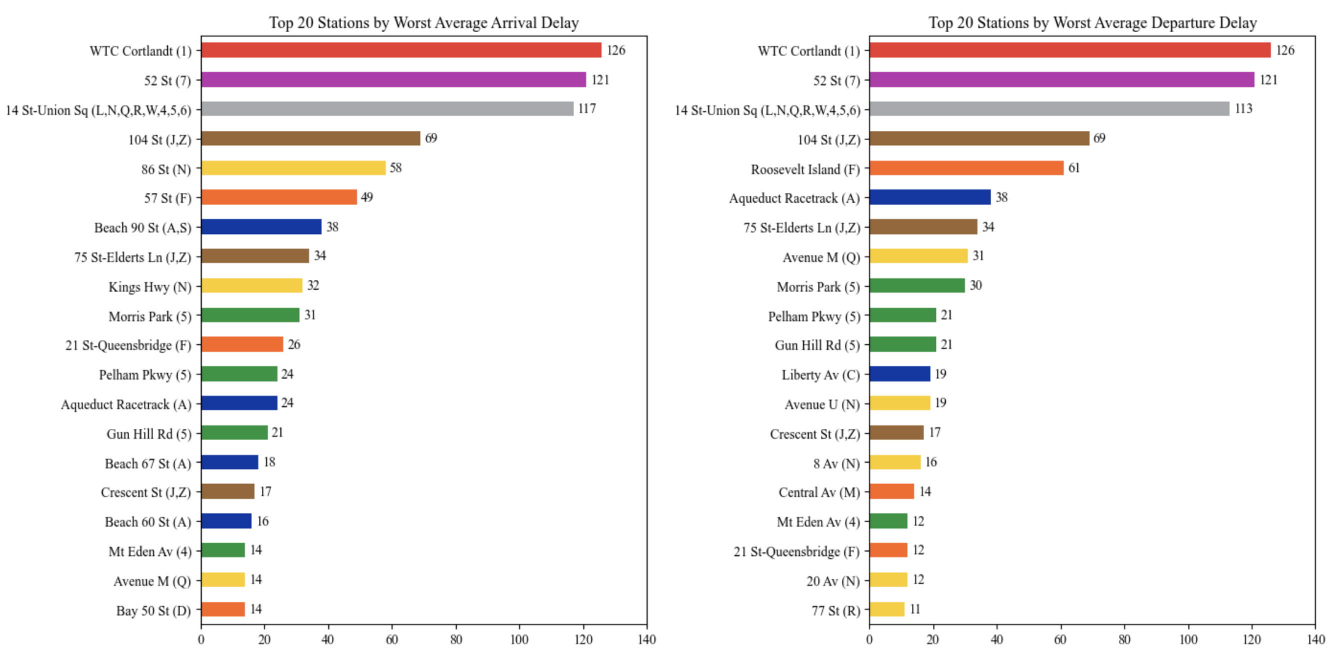


Fig. 9: Comparison of average arrival and departure delays of stations by line.

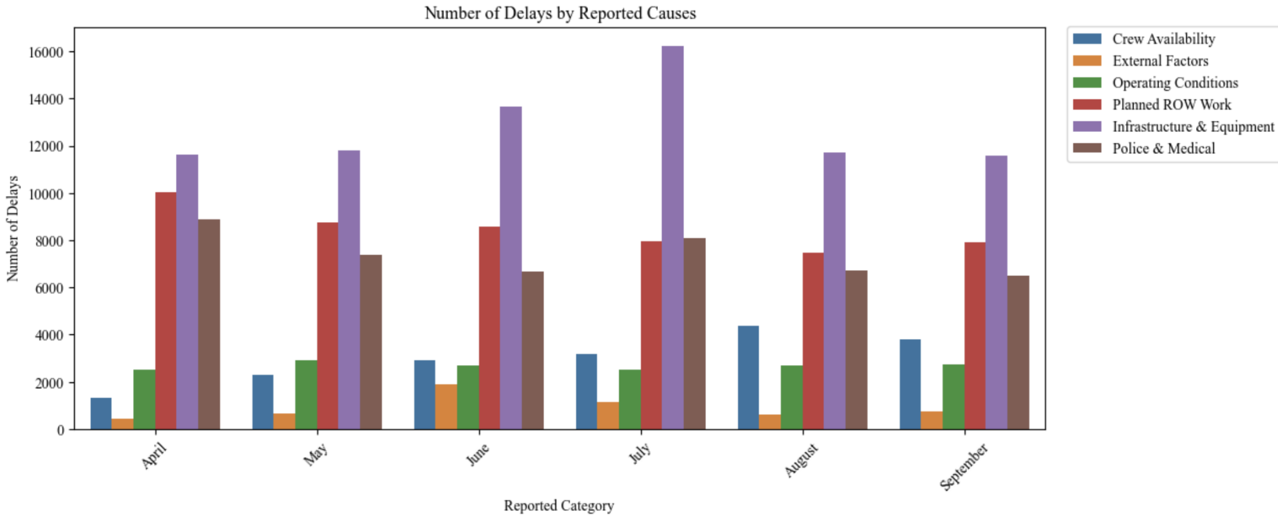


Fig. 10: Reported causes of delays.

Figure 10 shows the reported categories of delay causes from April to September 2025. Infrastructure & Equipment accounted for the largest share of reported delays across all months, followed by Planned Right-of-Way Work and Police & Medical. The elevated number of reported delays in June and July correspond with the increased delays shown in Figure 8, suggesting a possible seasonal effect. These seasonal increases may be associated with factors such as heat-related equipment stress, summer construction activity, or changes in travel patterns. While infrastructure-related issues dominate, the presence of police and medical incidents indicates that external, non-operational factors contribute meaningfully to service disruptions.

D. Connecting Crowding and Delay

To further examine the association between crowding and delay, each station was ranked by its total ridership, crowding index, average arrival delay, and average departure delay. Figure 11 presents four scatterplots comparing ridership rank and crowding rank against both average arrival and departure delays, with line-specific best-fit trendlines shown for each line color group.

The ridership-based comparisons suggest a modest inverse association between ridership and delays, as stations with higher ridership rankings tend to correspond with lower delay rankings and stations with lower ridership rankings often correspond with higher delay rankings. This pattern is reflected in the generally negative slopes observed for most line groups in both plots. Notably, the brown and gray lines differ from this broader pattern, exhibiting nearly horizontal trendlines which suggest weaker relationships between ridership and delays on those lines.

The crowding-based comparisons showed a more mixed pattern. While some line groups, particularly green, blue, and yellow, exhibit more negative relationships between crowding and delay ranks, several other line groups show comparatively flat trendlines. This suggests that the relationship between crowding and delays may vary substantially across line groups rather than following a consistent system-wide pattern.

A key result of this analysis is that the relationship between crowding and delay is not uniform across the subway system. While ridership may show a weak inverse association with delays overall, crowding appears to have a relationship with reliability that is more dependent on line groups. This is consistent with earlier findings that some lower-ridership lines, such as the brown line, experienced relatively high delays. This suggests that delay patterns may be shaped by more localized operational or infrastructure conditions than by passenger volume alone.

Taken together, these results suggest that crowding and delays may represent related but largely distinct dimensions of system strain, with their interaction varying across different parts of the network.

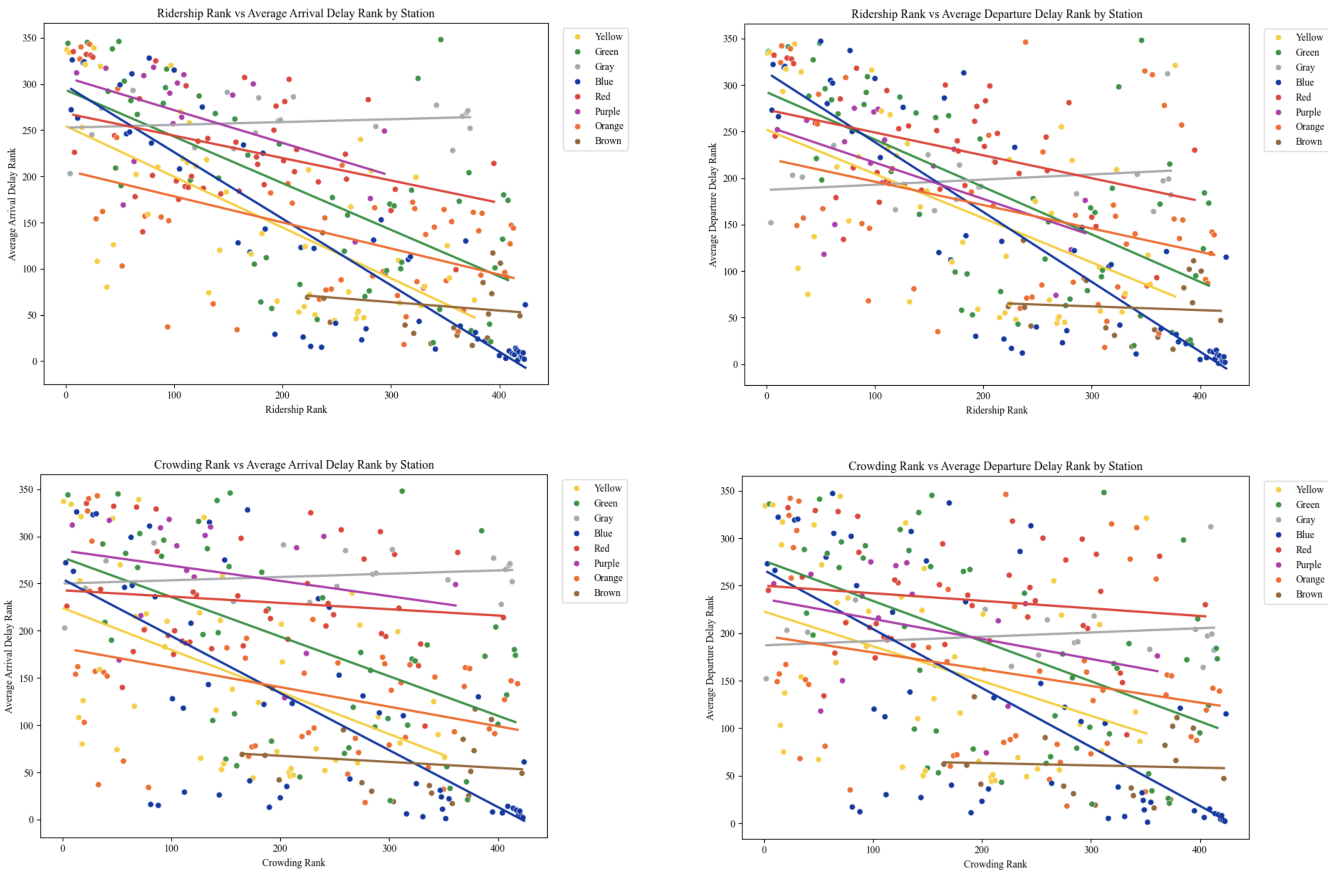


Fig. 11: Associations between station demand measures and reliability rankings.

V. Discussion

This analysis examined subway system performance through the dimensions of demand, represented through ridership and crowding patterns, and reliability, represented through measured delays and reported causes of service disruptions. The results suggest that crowding and delay capture related but distinct forms of system strain, as high ridership does not necessarily correspond to greater crowding pressure, and stations with the most severe delays often differ from the busiest stations. In addition, infrastructure and maintenance-related issues appeared as major contributors to station unreliability, suggesting that operational constraints may play a larger role in delay severity than passenger demand alone.

Several limitations should be considered when interpreting the results. First, ridership was used as a proxy for crowding, but ridership alone does not fully capture passenger density and platform congestion. Crowding can depend on other factors such as train frequency, vehicle capacity, station layout, and passenger distribution across time and space. Second, the delay analysis relied on matching scheduled GTFS stop times with real-time trip data using nearest-time alignment within a tolerance window. Though it is a practical method for estimating delays, it may introduce matching inaccuracies for closely-spaced trains, service irregularities, or complex operating conditions. Additionally, color-based line analysis was restricted to stations associated with a single line color to avoid ambiguity at multi-line stations, which excludes transfer-heavy locations from the analysis. Delay calculations focused on weekdays between 6 AM and midnight, intentionally excluding overnight service and weekends to maintain consistency. While this improves comparability, it limits conclusions about system performance outside of those periods. Finally, the reported categories for causes of delays identify broad causes of disruptions but do not directly measure the severity or duration of individual incidents, as well as incidents at specific times and locations.

Future work could incorporate train frequency and vehicle capacity data to better estimate crowding. Station-level measures or platform occupancy data would provide a more direct view of passenger congestion. Delay estimation could be refined through train-level or route-level matching instead of station-level matching, and it could incorporate multi-line transfer stations that were excluded from this study. Factors such as weather, special events, service changes, or infrastructure conditions could be included to further examine causes of delay.

VI. Conclusion

This project examined congestion and reliability challenges in the New York City subway system through the perspectives of ridership patterns and operational delay data. Using MTA ridership data, real-time trip data, GTFS schedules, and reported causes of delay data, this analysis explored how system strain varies across time, stations, and line groups.

The results showed that subway demand is strongly concentrated in peak commuting periods, while crowding analysis revealed patterns not captured by ridership alone. Using a crowding index and analysis of index distribution, the study highlighted differences in crowding intensity across stations, showing that some parts of the system experience greater demand pressure than others. The delay analysis also revealed differences in reliability across line groups and stations, with certain stations showing consistently higher average arrival and departure delays. In addition, reported delay causes demonstrated that infrastructure and equipment problems, as well as planned work and police and medical incidents, made up a large portion of disruptions, suggesting that the station's delays were tied to underlying system and maintenance challenges.

Taken together, these findings suggest that subway performance is influenced both by passenger demand and operational constraints. Crowding and delay should not be viewed as independent challenges, but rather as interconnected aspects of system strain. Furthermore, this project demonstrates how publicly available transit data can be used to analyze urban transportation performance and provide insight into where and when the subway system experiences the greatest pressure. Understanding these patterns is not only important for evaluating current system performance, but also for guiding future efforts to improve the resilience, reliability, and rider experience of the subway system.

Acknowledgment

I would like to thank Dr. Pramod Gupta for his mentorship and insightful feedback, which greatly contributed to the development and direction of this project.

References

[1] Subway and bus ridership for 2024. (2024). MTA. https://www.mta.info/agency/new-york-city-transit/subway-bus-ridership-2024

[2] Signaling of the New York City Subway. (2023, March 8). Wikipedia. https://en.wikipedia.org/wiki/Signaling_of_the_New_York_City_Subway

[3] MTA Dashboard. (2026). Mta.info. https://capitaldashboard.mta.info/budget-overview

[4] State of New York Open Data. 2026. MTA Subway Hourly Ridership: Beginning 2025 [Data set] .https://data.ny.gov/Transportation/MTA-Subway-Hourly-Ridership-Beginning-2025/5wq4-mkjj/data_preview

[5] Metropolitan Transportation Authority. MTA Developer Resources [Static GTFS Data]. https://www.mta.info/developers

[6] State of New York Open Data. 2026. MTA Subway Stations [Data set]. https://data.ny.gov/Transportation/MTA-Subway-Stations/39hk-dx4f/data_preview

[7] Subway Data NYC. Subway Data NYC [Data set]. https://subwaydata.nyc/

[8] Subway Data NYC. Subway Data NYC [Data set]. https://subwaydata.nyc/

[9] Caspari, A., Levine, B., Hanft, J., & Reddy, A. (2017). Real-Time Estimation of Platform Crowding for New York City Subway: Case Study at Wall Street Station on No. 2 and No. 3 Lines in Financial District. Transportation Research Record: Journal of the Transportation Research Board, 2648(1), 68–78. https://doi.org/10.3141/2648-08

[10] Ceapa, I., Smith, C., & Capra, L. (2012). Avoiding the crowds: understanding Tube station congestion patterns from trip data. Proceedings of the ACM SIGKDD International Workshop on Urban Computing - UrbComp '12. https://doi.org/10.1145/2346496.2346518

[11] Chu, K. K. A., & Chapleau, R. (2008). Enriching Archived Smart Card Transaction Data for Transit Demand Modeling. Transportation Research Record: Journal of the Transportation Research Board, 2063(1), 63–72. https://doi.org/10.3141/2063-08

[12] Daziano, R., & Nozick, L. K. (2017, November 30). Using Visual Information to Determine the Subjective Valuation of Public Space for Transportation: Application to Subway Crowding Costs in NYC. Bts.gov. https://rosap.ntl.bts.gov/view/dot/35387

[13] Douglas, N.J., & Karpouzis, G. (2006). Estimating the passenger cost of train overcrowding. https://australasiantransportresearchforum.org.au/wp-content/uploads/2022/03/2006_Douglas_Karpouzis_a.pdf

[14] Haywood, L., & Koning, M. (2015). The distribution of crowding costs in public transport: New evidence from Paris. Transportation Research Part A: Policy and Practice, 77, 182–201. https://doi.org/10.1016/j.tra.2015.04.005

[15] Haywood, L., Koning, M., & Monchambert, G. (2017). Crowding in public transport: Who cares and why? Transportation Research Part A: Policy and Practice, 100, 215–227. https://doi.org/10.1016/j.tra.2017.04.022

[16] Jang, W. (2010). Travel Time and Transfer Analysis Using Transit Smart Card Data. Transportation Research Record: Journal of the Transportation Research Board, 2144(1), 142–149. https://doi.org/10.3141/2144-16

[17] Lam, W. H. K., Cheung, C.-Y., & Lam, C. F. (1999). A study of crowding effects at the Hong Kong light rail transit stations. Transportation Research Part A: Policy and Practice, 33(5), 401–415. https://doi.org/10.1016/s0965-8564(98)00050-0

[18] Tirachini, A., Hensher, D. A., & Rose, J. M. (2013). Crowding in public transport systems: Effects on users, operation and implications for the estimation of demand. Transportation Research Part A: Policy and Practice, 53, 36–52. https://doi.org/10.1016/j.tra.2013.06.005